

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1456

Shiprock 2 100 kV
April 20, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1456. The Interconnection Customer proposes a 179.8 MW solar photovoltaic facility interconnecting at the Shiprock 2 100 kV substation in the Indiana area. The request was received on April 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 368 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$45,951,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Deers / Walsenburg 100 kV: 214.1% loading (98 MVA) following the outage of the Handsome Lake Energy Llc / Tamuning 100 kV Line; DFAX = 100.0%.

For the loss of Shiprock 2 / Walsenburg 100 kV, the Handsome Lake Energy Llc / Makahoy 100 kV loads to 203.4% of its 109 MVA rating. Project DFAX: 100.0%.

Shiprock 2 / Makahoy 100 kV: 199.6% loading (120 MVA) following the outage of the Shiprock 2 / Walsenburg 100 kV; DFAX = 100.0%.

Handsome Lake Energy Llc / Tamuning 100 kV: 196.1% loading (124 MVA) following the outage of the Walsenburg / Shiprock 2 100 kV; DFAX = 100.0%.

The Shiprock 2 / Walsenburg Substation 100 kV is overloaded to 192.8% of its 126 MVA emergency rating for the loss of the Tamuning / Handsome Lake Energy Llc 100 kV. The project contributes 100.0% DFAX (179.8 MW) to this overload.

For the loss of Handsome Lake Energy Llc / Tamuning 100 kV line, the Devers / Wrangell 100 kV Line loads to 171.5% of its 113 MVA rating. Project DFAX: 100.0%.

The Illinois Rd / Southport 100 kV is overloaded to 169.6% of its 127 MVA emergency rating for the loss of the Indego / Columbia Hills 100 kV L. The project contributes 6.7% DFAX (12.1 MW) to this overload.

The Southport / Hart 100 kV is overloaded to 156.7% of its 152 MVA emergency rating for the loss of the Indego / Columbia Hills 100 kV. The project contributes 6.7% DFAX (12.1 MW) to this overload.

Rmec Substation / Christopher 100 kV L: 156.5% loading (144 MVA) following the outage of the Napavine / S Haven 100 kV; DFAX = 22.4%.

The Indego / Waynedale 100 kV is overloaded to 150.5% of its 242 MVA emergency rating for the loss of the Monfort Substation / Waynedale 230 kV. The project contributes 10.2% DFAX (18.3 MW) to this overload.

The Hart / Portage Uppco 100 kV Line is overloaded to 147.6% of its 177 MVA emergency rating for the loss of the Indego / Columbia Hills 100 kV. The project contributes 6.7% DFAX (12.1 MW) to this overload.

Tamuning / Aladdin Substation 100 kV line: 143.5% loading (98 MVA) following the outage of the Shiprock 2 / Walsenburg 100 kV line; DFAX = 48.7%.

For the loss of Waynedale / Monfort 230 kV Line, the Indego / Colombia Hills 100 kV line loads to 135.4% of its 294 MVA rating. Project DFAX: 10.7%.

For the loss of Waynedale / Indego 100 kV, the Illinois Rd / Hebron Rosemeade 100 kV loads to 129.3% of its 163 MVA rating. Project DFAX: 8.0%.

For the loss of Rmec / Christopher 100 kV, the South Haven / Napavine Substation 100 kV loads to 116.8% of its 185 MVA rating. Project DFAX: 19.3%.

Aep Medford / Murdock 100 kV: 114.9% loading (165 MVA) following the outage of the Headwaters Wind Farm li Llc / Eaton 100 kV line; DFAX = 7.8%.

For the loss of Waynedale Substation / Indego 100 kV, the Hebron Rosemeade / Mkinley 100 kV loads to 105.2% of its 134 MVA rating. Project DFAX: 5.9%.

For the loss of Christopher / Rmec 100 kV L, the Murdock / Napavine 100 kV line loads to 103.0% of its 198 MVA rating. Project DFAX: 19.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Devers / Walsenburg 100 kV	Rebuild line with higher-capacity conductor	\$3,737,000	100% (\$3,737,000)
Makahoy / Handsome Lake Energy Llc 100 kV	Rebuild line with higher-capacity conductor	\$4,047,000	100% (\$4,047,000)
Shiprock 2 / Makahoy 100 kV	Rebuild line with higher-capacity conductor	\$8,717,000	100% (\$8,717,000)
Handsome Lake Energy Llc / Tamuning 100 kV	Rebuild line with higher-capacity conductor	\$3,402,000	100% (\$3,402,000)
Walsenburg / Shiprock 2 100 kV	Rebuild line with higher-capacity conductor	\$4,683,000	100% (\$4,683,000)
Devers / Wrangell 100 kV Line	Rebuild line with higher-capacity conductor	\$14,482,000	100% (\$14,482,000)
Illinois Rd / Southport 100 kV	Rebuild line with higher-capacity conductor	\$4,198,000	7% (\$291,000)
Southport / Hart 100 kV	Rebuild line with higher-capacity conductor	\$1,377,000	7% (\$96,000)
Rmec / Chistopher 100 kV Line	Rebuild line with higher-capacity conductor	\$4,249,000	16% (\$664,000)
Indego / Waynedale 100 kV	Rebuild line with higher-capacity conductor	\$4,726,000	7% (\$327,000)
Hart / Portage Uppco 100 kV Line	Rebuild line with higher-capacity conductor	\$4,039,000	7% (\$280,000)
Tamuning / Aladdin Substation 100 kV line	Rebuild line with higher-capacity conductor	\$4,484,000	73% (\$3,268,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Indego / Colmbia Hills 100 kV line	Rebuild line with higher-capacity conductor	\$1,953,000	7% (\$142,000)
Hebron Rosemeade / Illinois Road 100 kV	Rebuild line with higher-capacity conductor	\$2,511,000	7% (\$180,000)
Napavine / S Haven 100 kV	Rebuild line with higher-capacity conductor	\$4,663,000	14% (\$649,000)
Aep Medford / Murdock 100 kV	Rebuild line with higher-capacity conductor	\$1,312,000	17% (\$229,000)
Hebron Rosemeade / Mkinley 100 kV	Reconductor line	\$704,000	10% (\$69,000)
Murdock / Napavine 100 kV line	Reconductor line	\$4,943,000	14% (\$688,000)
	Total allocated to project		\$45,951,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.